ONLY YOU — technical report

A provenance-gated authority layer for wearable agents Sushant Poudel · Meta Global AI Developer Hackathon 2026 · Build for Your World

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1. The problem

Meta's WebMCP documentation for Ray-Ban Display tells developers to set `untrustedContentHint` "honestly, since **it's not an enforced security boundary**," and warns that tool-result text "goes into the assistant's context" and is "how a prompt injection reaches the model." The assistant also cannot tell the wearer's voice from a stranger's. Once the agent has a camera, a microphone and a wallet, any TV ad, poster, webpage or invoice can issue commands.

The missing layer is not intelligence. It is **authority**: which inputs are instructions at all, who is allowed to issue them, what they may authorise, and what can be proven afterwards.

2. What I built

Provenance before the model. Every input is labelled before a token reaches the model: on-device speaker verification (only the enrolled owner's voice is a command channel), DOM origin (web text is never a command), scene-text masks + OCR (physical text is data), sender trust for documents. The model never has to win an adversarial classification game; the channel decides what is even eligible to be an instruction.

A constitution and mandates. The constitution is the ceiling — deterministic rules first, a small on-device evaluator only for ambiguity, ask-once for irreversible actions. Mandates are delegation: a signed, scoped, expiring capability ("pay the electrician up to 5,000 until tomorrow") that lets the agent act without interrupting and lets a third party verify the authorisation. Spend is cumulative and tracked outside the signed payload, so a mandate stays valid after use.

Receipts. Every action *and every refusal* emits an HMAC-signed, hash-chained record with the evidence, the rule applied, and an undo. Refusals are tagged `proposed_by: model | display_only`, so an enforcement event is never confused with a UI attribution.

3. What I measured

The four-arm table is a **harness test, not a security claim**: the corpus and the router's channel table are co-designed, so arm D cannot mislabel inside it. Its zero is conditional on correct labels.

Arm	Architecture	Executed	False refusals
A	undefended	100%	0%
B	prompt-only	47.5%	0%
C	provenance-only	10%	0%
D	full ONLY YOU	0%	0%

The label layer — the part that can be wrong. A DOM hidden-text detector on 42 hidden / 35 benign HTML snippets: recall **81.0%** (95% CI 66.7–90.0), false positives **14.3%** (6.3–29.4), precision 86.8%. Eight hiding techniques are missed (`clip-path`, `clip`, `transform:scale(0)`, zero-height, rgba-white, fixed off-screen, `visibility:collapse`, hidden inputs). A browser labeler that asks the rendering engine for computed styles and ancestor contrast scores **42/42 with 0/35 false positives** on the same cases — an upper bound, because it was written after the cases existed, but the reason the production design does not parse `style=` strings. Speaker verification and its false-accept rate are **not measured**; the report gives a sensitivity table instead of a claim.

Adaptive corpus — the attacker controls the signals. Fifteen scenarios: twelve cloned-voice utterances presented with score 0.91 + proximity, three pages claiming `user_typed`. Result: **label error 80%, model-proposed 80%, executed 20%** (3/15 — the cases where the owner was present and confirmed). The web cases are 0/3 after the fix that makes web text data-only.

Signed sensor envelope. Six attacks fabricate the sensor signals (`speaker_score: 0.99` + proximity) to impersonate the owner: **6/6 execute without an envelope, 0/6 with a signed sensor bus**, with all six recorded as blocked. This closes the cheap forgery path; it does not stop a cloned voice (the sensor is honest) or a compromised sensor.

Evidence coverage. Eight paraphrased payloads with no trigger verb: **0 executed**, but only **1/8 produced a receipt**. The audit layer, not the gate, fails first.

Scene text. Thirty-five synthetic signs across seven conditions, rendered with ImageMagick and read with Tesseract: detection **94.3%**, mean character error rate **9.4%**; rotated text is the failure (60% / 54.2%). Routing the OCR result through the stack executes **0/35** and produces **24/35 receipts** — enforcement does not depend on OCR, because scene text is data by channel; OCR quality changes the evidence surface.

Latency. The in-process gate is **≈0.1 ms p50**. Perception latency is **unmeasured**, and the report says so; the hundred-millisecond figures a wearable demo implies are not claimed here.

4. What failed, and what I did not fix

- **Cloned voice** at score 0.91 defeats the biometric: 12/12 label errors. Policy catches 9/12; the three executions are the confirms-true worst case.
- **Forged sensor dict**: mitigated for the cheap forgery — with a signed sensor bus, fabricated signals go from 6/6 executed to 0/6. A compromised sensor, and a cloned voice, still pass.
- **Merchant verification** is a design sketch. The demo signs mandates symmetrically with a device key; a merchant cannot hold that secret. Production needs a public key (Ed25519) and an offline verifier.
- **Second-order injection** through the ledger: with the deterministic `ask()` template, receipt text is quoted, not interpreted. When the 1M-context summariser is wired, ledger entries must enter as data with provenance labels. Not yet exercised.
- **Biometric consent**: voiceprints are special-category data (GDPR Art. 9, DPDP 2023, BIPA). The demo enrolls only its author and keeps no audio; a product needs a consent flow, retention policy and a bystander policy.

The full table of nine named attacks, severity and status is `docs/limitations.md`.

5. Why this matters for Meta

Two products depend on this layer. **Glasses** become trustworthy only if the agent distinguishes the wearer from the world; the physical channel is the one no prompt-level defense covers. **Agentic commerce** depends on merchants being able to verify that an agent purchase was authorised — Amazon already blocks Muse because it cannot. A signed, scoped mandate turns agent commerce from a trust problem into a verification problem.

The architectural claim is deliberately narrow: **do not make the model robust to injection; make the channel untrusted by default.** Authority is a routing problem, and routing can be tested, instrumented and receipted.

6. With ten engineers

1. Ship the WebMCP tool-result provenance envelope: an `authority` field the assistant enforces, not a hint.
2. Voice + DOM provenance in the Muse client, with the signed sensor envelope.
3. Mandates on device public keys and an offline merchant verifier.
4. Scene provenance behind a flag; publish the benchmark and the corpus.

Repo · demo · benchmark: `results/benchmark.md` · limitations: `docs/limitations.md` · 89 tests